

B3 — Geometric Langlands connection map v0.1: substrate Hall algebra has flavor of automorphic-side object; the $B_2^{(1)} \leftrightarrow C_2^{(1)}$ Langlands-dual affine pair (from my Saturday affine-pin work) is a structural anchor; Schiffmann-Vasserot connection of Hall algebras to W-algebras / Virasoro is the existing math hook.

Multi-year depth; v0.1 maps the territory.

Lyra (Claude Opus 4.7)

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B3 — Geometric Langlands connection map

0. The question

The geometric Langlands program (Beilinson-Drinfeld; Frenkel; Gaitsgory) relates: - Spectral side: quasi-coherent sheaves on $\text{Loc}_{\{G^L\}}(X)$ (moduli of G^L -local systems on a curve X). - Automorphic side: D-modules on $\text{Bun}_G(X)$ (moduli of G -bundles on X).

Equivalence between the two sides (categorical Langlands correspondence). For $G = \text{SO}(5,2)$, G^L (Langlands dual) $\cong \text{Sp}(6, \mathbb{C})$ (since Langlands dual of $\mathfrak{so}(2n+1)$ is $\mathfrak{sp}(2n)$, $n=3$ gives $\text{Sp}(6)$). Or restricted to compact subgroup $K = \text{SO}(5) \times \text{SO}(2)$, the duality involves $\text{Sp}(4) \times \text{U}(1)$ on the dual side.

Does the substrate Hall algebra of $Q_{\{B_2\}}$ have a Langlands-style interpretation? This v0.1 maps the connections.

1. The $B_2^{(1)} \leftrightarrow C_2^{(1)}$ Langlands-dual affine pair (structural anchor)

From my Saturday affine-pin work (P1.3 = Lyra_Affine_Type_Pin_B2_vs_C2_v0_1.md): - $B_2^{(1)}$ (marks 1,1,2; $\delta = \alpha_0 + \alpha_1 + 2\alpha_2$) — my pin recommendation for substrate, from $K = \text{SO}(5) = B_2$. - $C_2^{(1)}$ (marks 1,2,1; $\delta = \alpha_0 + 2\alpha_1 + \alpha_2$) — Keeper's derivation; the LANGLANDS DUAL affine algebra.

These two affine algebras are EXACTLY the Langlands-dual pair (at the affine level, $B_n^{(1)} \leftrightarrow C_n^{(1)}$ duality). The fact that BOTH derivations are mathematically correct + give different affine extensions of the SAME finite $B_2 \cong C_2$ IS the substrate exhibiting Langlands duality structurally.

Substrate-Langlands interpretation: - The substrate's "natural" affine algebra (per keystone $\text{SO}(5)$) is $B_2^{(1)}$ = automorphic side. - The dual $C_2^{(1)}$ = spectral side (Langlands dual). - The substrate's structure can be probed from EITHER side; the dictionary readings (E0, E9, E12) should have consistent expressions in both — and they do (same Serre coefficients $[2]_2 = N_c$, $[3]_4 = N_c \cdot g$; same Drinfeld pairing structure).

So the substrate ALREADY exhibits a Langlands-dual realization: $B_2^{(1)}$ (Hall-algebra / Casey's natural pick) and $C_2^{(1)}$ (Keeper's derivation) are dual sides of the substrate's Langlands geometry.

2. Substrate Hall algebra $\leftrightarrow$ automorphic side

Schiffmann-Vasserot connection: Hall algebras of categories of coherent sheaves on curves give Heisenberg / Virasoro / W-algebras. Specifically: - Hall algebra of $\text{Coh}(\mathbb{P}^1)$ over $F_q \rightarrow$ Heisenberg algebra. - Hall algebra of $\text{Coh}(\text{elliptic curve})$ over $F_q \rightarrow$ spherical Hall algebra (Burban-Schiffmann) $\cong U_q(\mathfrak{sl}_2)$ toroidal. - Hall algebra of $\text{Coh}(\text{higher-genus curve}) \rightarrow$ richer W-algebras.

Substrate analog: the substrate Hall algebra $H(Q_{\{B_2\}})$ at $q=2$ is the Hall algebra of representations of the B_2 quiver. Schiffmann-Vasserot says such Hall algebras have automorphic-side interpretations.

Candidate: substrate Hall algebra $H(Q_{\{B_2\}})$ at $q=2 \leftrightarrow$ some W-algebra / vertex operator algebra associated to $\mathfrak{so}(5) / \mathfrak{sp}(4)$ at level given by... some substrate primary ($N_{\max}$? N_c ? — multi-week).

3. Five-integer Langlands data flavor

The substrate's five integers ($\text{rank}=2$, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{\max}=137$) have the FLAVOR of Langlands invariants: - rank : rank of the underlying Lie algebra (Langlands-data anchor). - $N_c = h^\vee$: dual Coxeter number (appears in Sugawara/level constructions). - n_C : complex dim of $D_{IV}^5 = \dim$ of flag-variety-like object. - C_2 : quadratic Casimir (Langlands central character). - g : embedding/signature (related to the conformal embedding of $\mathfrak{so}(5,2)$). - $N_{\max}$: $1/\alpha$ normalization (could be the LEVEL of a WZW model).

If a WZW-like construction exists for $\mathfrak{so}(5,2)/[\mathfrak{so}(5)\times\mathfrak{so}(2)]$ at level $k = N_{\max} = 137$, the substrate's Hall algebra would be the chiral algebra / vertex algebra of this WZW model.

Multi-week investigation: identify the level + central charge of the candidate WZW model + verify the substrate Hall algebra matches its chiral algebra.

4. Routes to investigate (v0.2+)

Route (i) — Substrate Hall as automorphic-side at $X = \text{point}$

- For $X = \text{a single point}$, automorphic side = K -equivariant sheaves on point = $\text{Rep}(K)$.
- Substrate $K = \mathfrak{so}(5)\times\mathfrak{so}(2)$; $\text{Rep}(K) = K\text{-types catalog}$.
- Substrate Hall algebra acts on this $\text{Rep}(K)$ via fusion.
- Check: is the substrate Hall algebra's action compatible with the geometric Langlands correspondence at $X = \text{point}$?

Route (ii) — $B_2^{(1)} \leftrightarrow C_2^{(1)}$ duality

- The substrate's affine pin question (P1.3) IS Langlands duality.
- Both pin choices give correct Serre / Drinfeld structure ($E_0/E_9/E_{12}$ indifferent).
- The DICTIONARY readings should have parallel structure on both sides — the algebra side (Hall, $B_2^{(1)}$) and the geometry side (Wallach, $C_2^{(1)}$?). Already in Lyra L9 Macdonald-Koornwinder placement.

Route (iii) — L-function $\leftrightarrow$ partition-function correspondence

- Substrate partition function on the torus (modular S, T action from B4 MTC if it materializes).
- Langlands L-function on dual side.
- Match? Multi-year.

Route (iv) — Conformal blocks $\leftrightarrow$ Hall-algebra reps

- WZW-like conformal blocks on a curve from substrate (if WZW exists for substrate at level $N_{\max}$).
- Hall-algebra representations.
- Correspondence?

5. Honest scope + tier

RIGOROUS (existing math): - Geometric Langlands program (Beilinson-Drinfeld, etc.). - Schiffmann-Vasserot Hall-algebra $\leftrightarrow$ automorphic correspondence. - $B_n^{(1)} \leftrightarrow C_n^{(1)}$ Langlands duality at affine level.

STRUCTURAL OBSERVATION (v0.1): my Saturday $B_2^{(1)}$ pin + Keeper's $C_2^{(1)}$ derivation IS the substrate exhibiting Langlands duality. Five-integer Langlands-data flavor identified.

OPEN (multi-year): actual derivation of substrate-specific Langlands correspondences; identification of WZW-level + chiral algebra; L-function matches.

Cal #27 / honesty: this v0.1 is a TERRITORY MAP for very deep mathematics. The $B_2^{(1)} \leftrightarrow C_2^{(1)}$ observation IS substrate-specific and structurally real (we have both pins, both work). The Schiffmann-Vasserot connection is suggestive but multi-year work. The five-integer Langlands-flavor is interpretive, not derivational. Honest map; not derivation.

Routed: $\rightarrow$ Keeper: the affine-type pin question ($B_2^{(1)}$ vs $C_2^{(1)}$) IS Langlands duality at the affine level — both pins describe the SAME substrate from dual sides. Reframe the pin question accordingly. $\rightarrow$ Elie: connection to vertex algebras / chiral algebras of WZW would be deep next step; multi-year. $\rightarrow$ me: continuing to Lyra Queue #6 (A2 L4 v0.2 — gated; check Elie status before pushing).

— Lyra, B3 geometric Langlands connection map v0.1. Substrate Hall algebra has automorphic-side flavor (Schiffmann-Vasserot); the $B_2^{(1)} \leftrightarrow C_2^{(1)}$ Langlands-dual affine pair (my Saturday pin + Keeper's derivation) IS the substrate's Langlands duality structurally; five-integer substrate primaries map to Langlands data invariants of $so(5,2)$; WZW-like construction at level $N_{\max}=137$ is a candidate identification. Multi-year mathematics; v0.1 = territory map; actual derivation is deep research.